

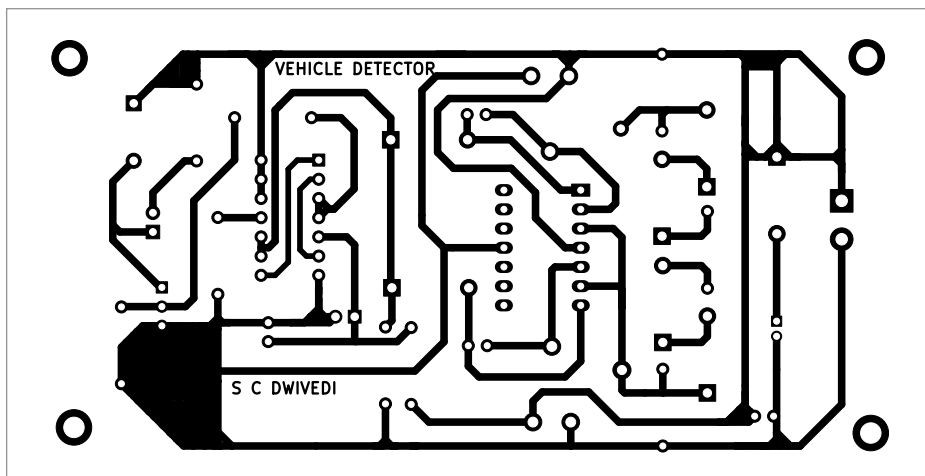


Fig. 3: Actual-size PCB

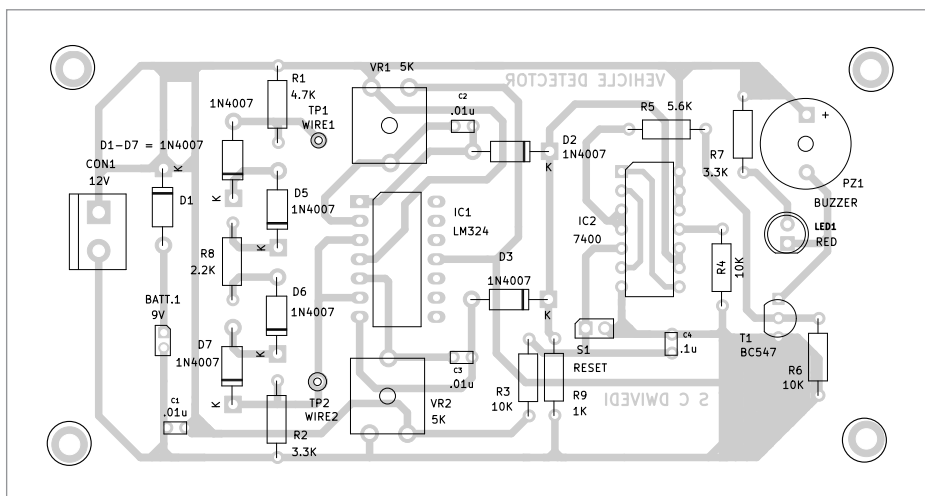


Fig. 4: Components layout

- Repeat the tests using a 9V battery, disconnecting the 12V DC adaptor

After satisfactory testing, insulate and seal the sensor circuit in a small tube with epoxy compound, leaving only the thick leads exposed for external connection. The internal circuit of the sensor should not be visible to anyone.

Mount the sensor on the vehicle and connect it to the alarm circuit using a suitable length of weather-resistant twin-core cable. The connection near the vehicle can be done with crocodile clips, an auto wiring plug and socket, or any

other connector type, based on user preference.

How it protects your vehicle from theft

The main circuit, including the assembled PCB, should remain inside the house. The vehicle to be protected should be fixed with sensor and twin wire and connected between TP1 and TP2. If someone attempts to steal the vehicle, the wire will break, triggering the buzzer to alert the user.

Caution

The 12V power supply adaptor must be of the mains-isolated type, either

step-down transformer and rectifier-based or SMPS type. Non-isolated power supplies with capacitors or resistors for voltage reduction must never be used, as this could lead to electric shock or electrocution when connecting or disconnecting wires to the vehicle sensor.

To ensure trouble-free operation, the connecting wires should be laid carefully so that they are not disturbed by nearby people, children, or animals.

The device has been tested with a 20-metre-long, 1mm dia, single-core cable (two wires), as shown in the photo. However, such a thick cable is unnecessary. It is recommended to use a twin-core copper flat cable (23/0052, 1100V grade).

EFY note. It would be better to assemble D4, D5, R8, D6, and D7 on a zero PCB rather than a PCB shown in Fig. 4. This assembly functions as a sensor. Fix it in the vehicle and connect the two ends of the sensor wires (from the vehicle) to the terminals TP1 and TP2 of the main

circuit using two-core long wires.

Modification

If a louder hooter is required, a 12V sugar cube relay with a freewheeling diode can be connected in parallel to the existing buzzer. The hooter can then be connected through the relay's normally open (NO) contact, powered by the 12V supply. **EFY**



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